

S27- Bayes' Theorem & Business Probability

Instructions: Solve all questions in Excel. Show calculations wherever required. Focus on **calculation + interpretation + business decision**, not formula memorization.

1 | Bayes' Theorem — Customer Conversion

An e-commerce company has two traffic sources:

Traffic Source	Visitors	Purchased
Advertisement	600	90
Organic Search	400	100

Answer:

1. What is the total number of visitors?
 2. What is the total number of purchasers?
 3. What is $P(\text{Advertisement})$?
 4. What is $P(\text{Purchased} \mid \text{Advertisement})$?
 5. What is $P(\text{Purchased})$?
 6. If a randomly selected customer purchased, what is the probability that they came from Advertisement?
 7. Which traffic source has the higher purchase rate?
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2 | Reverse Probability

A company receives applications from two sources:

Source	Applications	Selected
LinkedIn	500	75
Job Portal	300	30

Answer:

1. Total applications.
 2. Total selected candidates.
 3. $P(\text{Selected})$.
 4. $P(\text{Selected} \mid \text{LinkedIn})$.
 5. $P(\text{LinkedIn} \mid \text{Selected})$.
 6. Is $P(\text{Selected} \mid \text{LinkedIn})$ the same as $P(\text{LinkedIn} \mid \text{Selected})$?
 7. Which probability would be useful if HR wants to know **where a selected candidate came from**?
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3 | Bayes — Product Return Investigation

An online store has two product categories:

Category	Orders	Returned
Electronics	800	80
Clothing	1200	60

Answer:

1. Total orders.
2. Total returned orders.
3. Probability that an order is returned.
4. Probability that an Electronics order is returned.

5. Probability that a returned order was Electronics.
 6. If an order is known to be returned, which category is more likely?
 7. Does having a higher number of orders automatically mean having a higher return probability?
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4 | Conditional Probability vs Reverse Conditional Probability

A food delivery company records:

Customer Type	Customers	Ordered Again
New Customer	700	140
Existing Customer	300	120

Answer:

1. Total customers.
 2. Total customers who ordered again.
 3. $P(\text{Ordered Again} \mid \text{New Customer})$.
 4. $P(\text{Ordered Again} \mid \text{Existing Customer})$.
 5. $P(\text{Existing Customer} \mid \text{Ordered Again})$.
 6. Which customer type has the higher repeat-order rate?
 7. Which probability would answer: **“Among repeat customers, how many were existing customers?”**
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5 | Bayes — Fraud Detection

A bank analyzes transactions:

Transaction Type	Transactions	Fraud
Online	7,000	140
Offline	3,000	30

Answer:

1. Total transactions.
 2. Total fraudulent transactions.
 3. Overall fraud probability.
 4. $P(\text{Fraud} \mid \text{Online})$.
 5. $P(\text{Online} \mid \text{Fraud})$.
 6. If a transaction is known to be fraudulent, which type is more likely?
 7. Can we conclude that every online transaction is risky?
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6 | Business Decision — Marketing Campaign

A company runs two campaigns:

Campaign	Customers Reached	Customers Purchased
Campaign A	2,000	160
Campaign B	1,000	120

Answer:

1. Total customers reached.
 2. Total purchases.
 3. Purchase rate for Campaign A.
 4. Purchase rate for Campaign B.
 5. Probability that a randomly selected purchaser came from Campaign A.
 6. Which campaign has the better conversion rate?
 7. Which campaign generated more purchasers?
 8. Are **conversion rate** and **number of purchasers** measuring the same thing?
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7 | Bayes — Employee Skill Analysis

A company has employees from two departments:

Department	Employees	Python Skilled
Data Analytics	80	60
Marketing	120	24

Answer:

1. Total employees.
 2. Total Python-skilled employees.
 3. $P(\text{Python Skilled})$.
 4. $P(\text{Python Skilled} \mid \text{Data Analytics})$.
 5. $P(\text{Data Analytics} \mid \text{Python Skilled})$.
 6. If an employee is Python-skilled, which department are they more likely to belong to?
 7. Is the answer to Q4 the same as Q5?
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8 | Find the Wrong Interpretation

A company finds:

$$P(\text{Purchase} \mid \text{Advertisement}) = 30\%$$

An analyst makes these statements:

1. 30% of advertisement viewers purchased.
2. Every 100 advertisement viewers will definitely produce 30 purchases.
3. Advertisement viewers have a 30% observed purchase probability.
4. The probability describes likelihood, not a guaranteed future result.
5. We can automatically conclude that the advertisement **caused** the purchases.

Identify **all incorrect interpretations** and explain why.

9 | Bayes — Customer Support

A company receives support requests through two channels:

Channel	Requests	Resolved
Chat	600	480
Email	400	240

Answer:

1. Total requests.
 2. Total resolved requests.
 3. $P(\text{Resolved})$.
 4. $P(\text{Resolved} \mid \text{Chat})$.
 5. $P(\text{Resolved} \mid \text{Email})$.
 6. $P(\text{Chat} \mid \text{Resolved})$.
 7. Which channel has the higher resolution rate?
 8. If a randomly selected request is known to be resolved, is it more likely to have come from Chat or Email?
 9. Give one business recommendation based on the results.
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10 | Full Data Analyst Case — Customer Churn

A subscription company analyzes customer churn:

Customer Type	Customers	Churned
Monthly Plan	1,500	300
Annual Plan	2,500	125

Answer:

1. Calculate the total number of customers.
2. Calculate the total number of churned customers.

3. Calculate the overall churn probability.
4. Calculate $P(\text{Churn} \mid \text{Monthly Plan})$.
5. Calculate $P(\text{Churn} \mid \text{Annual Plan})$.
6. Calculate $P(\text{Monthly Plan} \mid \text{Churn})$.
7. Which plan has the higher churn rate?
8. Among churned customers, which plan contributes more customers?
9. Does the plan with more churned customers necessarily have the higher churn rate?
10. If you were a Data Analyst, which plan would you investigate first and **why**?
11. Would you recommend immediately removing the Monthly Plan based only on this data?
12. What additional information would you want before making a final business decision?